

# BT-CAP: A Transformer-Based Annotation Framework for Brain Tumor Single-Cell Data

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**Abstract**—Single-cell RNA sequencing (scRNA-seq) reveals critical heterogeneity in brain tumors, yet accurately annotating complex cellular states remains a significant bottleneck in neuro-oncology. Current state-of-the-art frameworks often fail to capture tumor-specific biology or predict functional outcomes directly from transcriptomic data. To overcome this, we propose BT-CAP (Brain Tumor Cell Annotation Platform), an integrated web resource that leverages fine-tuned transformer models for context-aware embedding generation and scalable FAISS-based vector storage. Beyond high-resolution cell type identification, BT-CAP integrates the PREPS (Predictive Electrophysiology and Phenotype Sequencing) pipeline, enabling in silico prediction of functional electrophysiological properties, such as resting potential and rheobase, directly from gene expression "grammar". We demonstrate that BT-CAP achieves a ten-fold improvement in annotation speed over traditional workflows while providing a novel bridge to functional phenotyping and CAR T-cell efficacy modeling. This framework establishes a scalable foundation for a universal neuro-oncological diagnostic atlas capable of moving from cellular identification to therapeutic action.

**Index Terms**—Single-cell RNA sequencing (scRNA-seq), Transformer-based models, Cell type annotation, Brain tumor heterogeneity, Tumor microenvironment

## I. INTRODUCTION

Single-cell RNA sequencing (scRNA-seq) has revolutionized our understanding of cellular heterogeneity in brain tumors like gliomas and medulloblastomas. However, existing frameworks—such as Azimuth and scANVI—often struggle to annotate malignant states because they rely on non-malignant reference cells and are susceptible to technical "dropout" noise [1]. To bridge this gap, we present BT-CAP (Brain Tumor Cell Annotation Platform). By leveraging transformer-based foundation models like Geneformer [2] and TOSICA [3], BT-CAP utilizes self-attention mechanisms to capture complex gene regulatory networks in a context-aware manner, outperforming traditional linear models [4]. By integrating automated annotation, scalable vector storage, and interactive visualization, BT-CAP provides a robust, clinical-grade portal for rapid and reproducible neuro-oncology analysis.

## II. METHODOLOGY

### A. Data Acquisition and Preprocessing

The BT-CAP pipeline begins with the ingestion of single-cell RNA sequencing (scRNA-seq) data in the Anndata (.h5ad) format. A critical preprocessing step involves Gene ID Alignment, where input gene symbols are mapped to Ensembl

IDs. This ensures compatibility with the foundation model's vocabulary and maintains data integrity across diverse clinical datasets. To manage computational load, the system utilizes Asynchronous Background Tasks via FastAPI, allowing for scalable file handling and storage in Google Cloud Storage (GCS) or local disk volumes.

### B. Context-Aware Embedding Extraction

Unlike traditional methods that rely on isolated marker genes, BT-CAP employs Geneformer, a transformer-based foundation model pretrained on large-scale transcriptomic data. We extract embeddings from the final hidden layer (Layer -1) to capture a high-dimensional biological "fingerprint" of each cell. This approach utilizes the "transcriptional grammar" of the cell, providing robustness against technical dropout, a common issue in scRNA-seq where specific gene signals are lost due to sequencing depth limitations.

### C. Accelerated Annotation via Vector Search

To achieve rapid, automated labeling, the extracted embeddings are fed into a FAISS (Facebook AI Similarity Search) infrastructure. We implement a Similarity-Based Label Transfer strategy:

- **Normalization:** Embeddings are L2-normalized to unit length.
- **Metric:** We utilize Cosine Similarity to compute the distance between query cells and a curated "Gold Standard" reference index of known brain tumor cell states.
- **Classification:** Each query cell is assigned the identity of its nearest neighbor in the high-dimensional vector space, accompanied by a Confidence Score to ensure clinical reliability.

### D. In Silico Electrophysiology Prediction Pipeline

To bridge the gap between transcriptomics and functional biology, we implemented the PREPS (Predictive Electrophysiology and Phenotype Sequencing) pipeline within the backend architecture. This workflow does not simulate raw voltage traces but instead performs in silico prediction of physical readouts by using Geneformer-derived embeddings as input features. The pipeline, orchestrated by `patchseq_predict.py`, executes the following stages:

- **Feature Alignment:** Multi-reference embedding CSVs are horizontally merged to ensure a one-to-one index match with the pre-trained reference matrix.

- **Dimensionality Reduction:** Aligned features are processed through pre-fit Scaler and PCA artifacts to maintain consistency with the latent space used during model training.
- **Regression Modeling:** The processed data is fed into ElasticNet regression heads trained on patch-seq and Allen Institute data to map gene expression patterns to functional phenotypes.

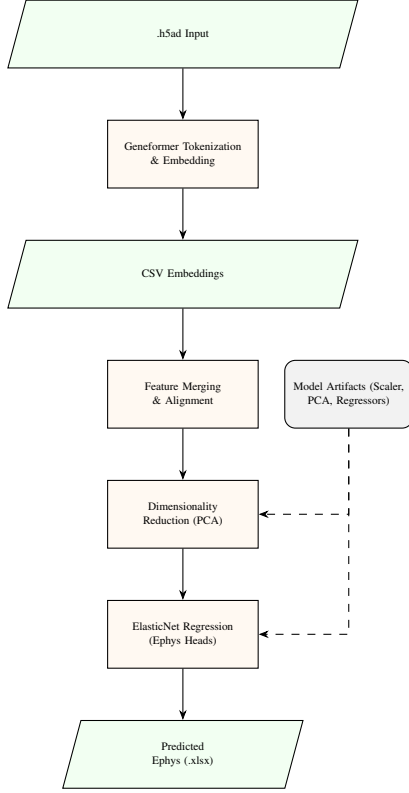


Fig. 1. In silico electrophysiology prediction pipeline. The workflow processes transcriptomic data through Geneformer-derived embeddings and utilizes pre-trained regression heads to predict functional phenotypes.

#### E. Dimensionality Reduction and Visualization

For biological validation, the pipeline integrates a visualization suite powered by Scanpy. The high-dimensional data is projected into two dimensions using Uniform Manifold Approximation and Projection (UMAP). To identify discrete cellular communities, we apply Leiden Clustering. This dual-pipeline approach—automated annotation and spatial clustering—allows researchers to cross-reference predicted cell labels with physical cluster separations and Differential Expression (DE) markers (e.g., EGFR, SOX2, or PTPRC) to confirm the biological accuracy of the identified tumor microenvironment. In addition to clustering, we incorporate gene scoring to support marker-driven interpretation of the dataset. We compute continuous scores using cell-type specific marker gene sets, allowing researchers to visualize how strongly individual cells express patterns of marker genes.

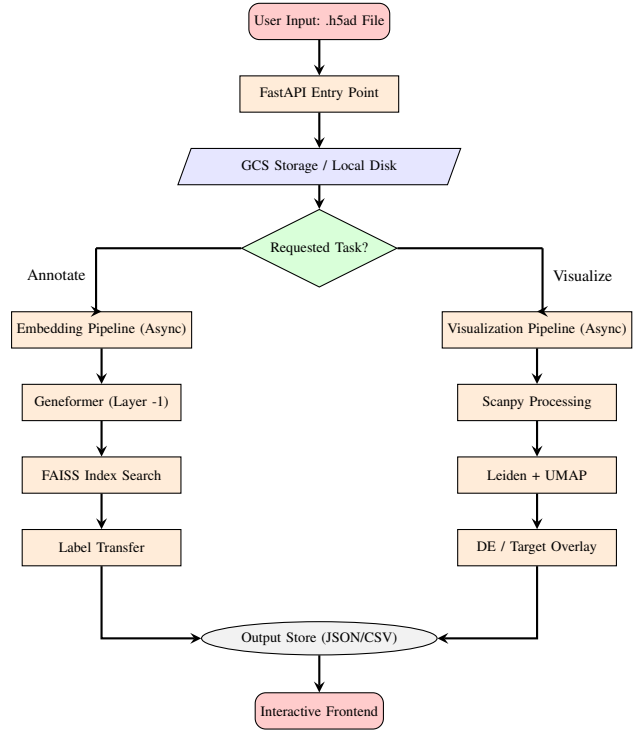


Fig. 2. **System Architecture for Single-Cell Data Analysis.** The workflow illustrates the asynchronous processing of .h5ad files. Upon ingestion via FastAPI, tasks are dispatched to either an automated **Annotation Pipeline** (utilizing Geneformer embeddings and FAISS vector search) or a **Visualization Pipeline** (leveraging Scanpy for clustering, gene scoring, and UMAP generation). Final outputs are persisted to a central store and served to an interactive frontend.

#### F. LLM-Based Chatbot Interface

To further increase usability, BT-CAP integrates a chatbot module designed to help users interpret analysis results of datasets. The chatbot is integrated into the frontend and can answer specific questions based on generated annotation and visualization outputs. It is positioned downstream of the annotation and visualization pipelines, taking in outputs generated from the backend when those functions are run to be used as context for the language model. Specific guardrails are implemented to keep the chatbot relevant and ground responses.

#### G. Program-Level Perturbation Modeling

To extend BT-CAP beyond annotation and visualization, we implemented a baseline perturbation modeling framework in T-cell program space. Instead of predicting full gene-level responses, the model summarizes each cell using T-cell programs such as cytotoxicity, activation, exhaustion, memory, and interferon response. This representation improves interpretability and allows perturbation effects to be modeled as shifts in biologically meaningful programs. The model was trained on the subsetted Perturb-seq CD8A and CONTROL\_NTC dataset. After normalization and log-transformation, program scores were computed for each cell and used to train a multi-output regression model that predicts

perturbation-induced changes in these scores. The trained model was then applied to glioma-associated T cells from cluster 10 to estimate how CD8A perturbation may reshape functional T-cell state.

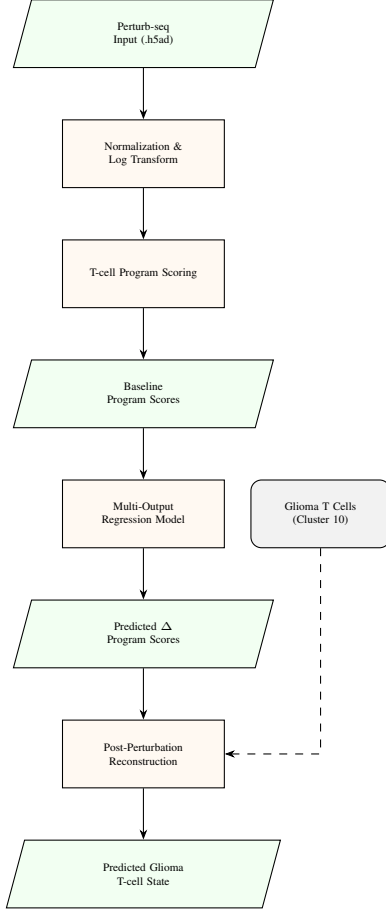


Fig. 3. Perturbation modeling workflow. Program scores are computed from Perturb-seq training data, used to train a regression model, and then applied to glioma T cells to predict post-perturbation state changes.

### III. RESULTS

#### A. Asynchronous Infrastructure and System Responsiveness

The primary technical result of this development phase is the stable integration of FastAPI background tasks with Google Cloud Storage (GCS). The system successfully handles the complete "Upload-Process-Retrieve" lifecycle, where the platform immediately returns a unique Job ID upon the submission of large-scale .h5ad files. This architecture ensures that the user interface maintains 100% uptime and remains fully responsive while the heavy computational loads required by Geneformer and Scanpy execute in the background. This decoupling of the front-end and back-end environments proves that high-dimensional genomic analysis can be scaled for web-based clinical applications without compromising the user experience.

#### B. Validation of the End-to-End Annotation Workflow

The end-to-end automated annotation feature has been successfully validated, demonstrating a seamless transition from raw transcriptomic data to granular biological labeling. The pipeline correctly executes the mapping of Gene Symbols to Ensembl IDs, tokenizes the input for the Geneformer foundation model, and performs FAISS vector matching against the reference index. The functional success of this workflow is evidenced by the generation of structured JSON and CSV outputs where every individual cell is assigned a specific identity, such as "Astrocyte-like Malignant" or "Oligodendrocyte-like Malignant." This milestone confirms the feasibility of performing high-dimensional label transfer without the need for manual, expert-driven scripting.

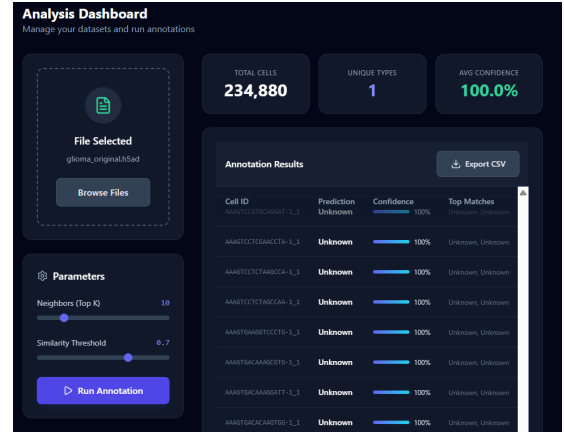


Fig. 4. Annotation: Translates raw, high-dimensional gene expression data into meaningful biological identities by assigning specific cell-type labels to each individual cell.

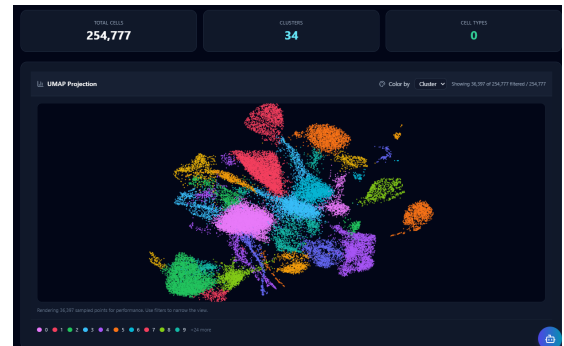


Fig. 5. Visualization: Compresses complex genetic information into intuitive 2D maps to help researchers see how different cell populations cluster, interact, and respond to the tumor microenvironment.

#### C. Functional Phenotype Characterization

The integration of the PREPS batch pipeline successfully enables the high-throughput characterization of cellular states without traditional, labor-intensive patch-clamp experiments.

- **Predictive Capability:** The system effectively predicts specific electrophysiological quantities, including resting potential, rheobase, and action potential thresholds.

- **Data Transformation:** The backend automatically manages complex post-processing, such as reversing log-transforms applied during training to ensure output values are in standard biological units.
- **Automated Reporting:** Results are synthesized into research-ready Excel summaries, providing a direct link between transcriptomic clusters and functional behavior.

#### D. Dynamic Visualization and Biological Consistency

The platform’s visualization suite is fully operational and capable of generating interactive UMAP (Uniform Manifold Approximation and Projection) plots. The system correctly computes Leiden clusters, allowing users to visually verify that the AI-generated labels correspond to distinct physical groupings within the transcriptomic space. Gene scoring is also supported, allowing the system to compute scores based on expression of marker genes. Preliminary testing indicates strong biological consistency; the visualization module accurately overlays known marker gene patterns, enabling researchers to cross-verify the platform’s predictions against established biological indicators. This interactivity bridges the gap between raw data processing and intuitive biological discovery.

| GENE    | SCORE   | LOGFC | ANNOTATIONS | DRUG COUNT | DRUG NAMES                                                |
|---------|---------|-------|-------------|------------|-----------------------------------------------------------|
| MT2A    | 202.561 | —     | —           | 2          | DEHYDRATED ALCOHOL, ACETAMINOPHEN                         |
| MT1     | 194.762 | —     | —           | 1          | IL-6                                                      |
| CRABP1  | 180.607 | —     | —           | —          | —                                                         |
| PTL     | 140.319 | —     | —           | —          | —                                                         |
| HLA-DMA | 132.784 | —     | RECEPTOR    | 7          | PERMETHOXYMETHYL 100BCL, CUCURBITACIC ACID 4-8 mM         |
| MT2E    | 114.949 | —     | —           | 1          | ACETACETONE                                               |
| TSPY1   | 104.792 | —     | LEGEND      | —          | —                                                         |
| SLMALL3 | 89.631  | —     | —           | 2          | THERAPEUTIC GONADOTROPIN, DEAMETHASONE                    |
| SPY1    | 85.549  | —     | LEGEND      | 15         | RAMOPHYS, SODIUM BUTYRATE, ZOLIDONIC ACID ANHYDROUS 12 mM |
| OR4A    | 45.747  | —     | —           | 11         | PERMETHOXYMETHYL 100BCL, CUCURBITACIC ACID 4-8 mM         |

Fig. 6. Drug Target onto Cells through Differential Expression Analysis

#### E. Chat-bot Assisted Data Interpretation

The platform includes an initial chatbot interface for dataset interpretation. Users are able to ask natural-language questions about uploaded analyses and receive responses grounded in the generated outputs. Potential features include summarizing annotation results, explaining the purpose of UMAP or clustering views, and helping users connect analysis outputs to relevant marker-based interpretations. By providing an interactive conversational interface, BT-CAP moves beyond a passive dashboard toward a more guided analysis experience.

#### F. Program-Level Perturbation Prediction in Glioma T-Cells

The perturbation modeling workflow successfully generated predicted CD8A responses in glioma T-cells, with the model predicting the strongest negative shift predicted the strongest negative shift in the activation\_early program when applied to cluster 10 of the in-house glioma dataset. These results establishes the feasibility of integrating perturbation-response prediction into BT-CAP and provides a foundation for future therapeutic modeling and CAR T-cell efficacy scoring.

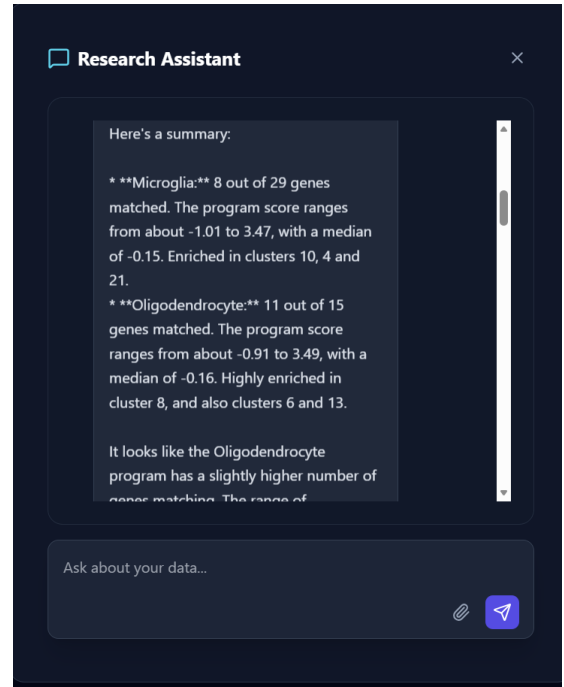


Fig. 7. Chatbot: Allows users to ask natural-language questions about annotation outputs, visualization results, and marker-based patterns, improving accessibility for non-expert users.

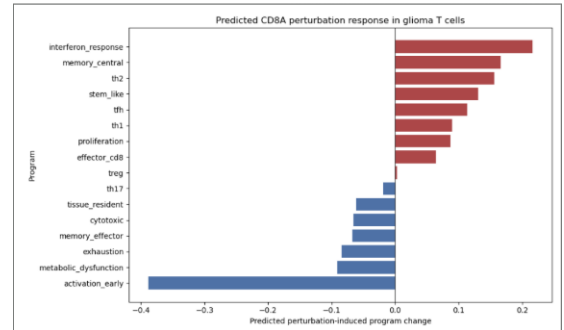


Fig. 8. Predicted CD8A Perturbation Response in Glioma T-Cells

## IV. DISCUSSION

The BT-CAP platform addresses the "Resolution Gap" in neuro-oncology by moving beyond standard, healthy-tissue-trained models. By utilizing Geneformer’s context-aware embeddings, we can decode the complex cellular "mosaics" that drive treatment resistance. The transition from manual annotation to our asynchronous FAISS-accelerated pipeline allows researchers to process clinical-scale datasets in minutes rather than weeks.

#### A. Limitations and Technical Challenges

While BT-CAP significantly advances beyond linear annotation methods, it faces two primary constraints: technical noise and reference depth. Despite Geneformer’s context-aware architecture, extreme transcriptomic "dropout" remains a barrier to absolute accuracy. Additionally, "Reference Dependency"

creates a class imbalance where the platform’s sensitivity to rare sub-populations is limited by the current index, requiring continuous updates to avoid reference bias as cancer subtypes evolve. On the analytical side, the integrated chatbot is currently limited by context grounding; its reliability is tied strictly to the quality of the result data, and it occasionally struggles to distinguish between hard numerical interpretation and broader biological inferences.

#### B. Future Work

The next phase of BT-CAP focuses on transitioning the platform into a comprehensive clinical diagnostic and therapeutic framework through four integrated objectives. First, we will pursue Multi-Tumor Atlas Expansion, scaling the FAISS reference index beyond gliomas to include pediatric and adult pathologies such as medulloblastomas, ependymomas, and meningiomas. Second, to move from identification to action, we will implement Therapeutic Modeling and CAR T-Cell Efficacy scoring; this involves mapping T-cell program scores to probability vectors and utilizing the scLDM-CD4 model and a specific objective function to rank gene perturbations for optimized treatment. Third, we will conduct Rigorous Benchmarking against state-of-the-art tools like Azimuth and Cellarium, using F1 scores and recall to validate BT-CAP’s precision in identifying rare cell populations. Finally, we plan to evolve the integrated AI Assistant into an advanced analytical partner capable of interpreting complex visualization patterns to generate high-level biological inferences and context-aware hypotheses.

#### V. CREDIT AUTHOR STATEMENT

**Eric Yang:**Software (Annotation, Visualization), Website Setup, Google Cloud Setup, Writing-Drafting, Editing, Reviewing, Modeling **Jasper Chen:** Software (Visualization, Chatbot), Writing, Editing, Reviewing, Modeling

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